



Optical property evaluation of oxyfluoride glasses doped with different amounts of Y^{3+} ions

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ABSTRACT

In present study, oxyfluoride glasses doped with different amounts of Y_2O_3 (0.5, 1 and 1.5 wt.%) were prepared by the convenient melting process. Based on the density measurements and differential thermal analysis (DTA) results, the network-forming role of Y^{3+} ions was proved. In addition, Fourier transform infrared spectroscopy (FT-IR) study of the samples approved this claim by showing structural changes happening with the increase of Y_2O_3 content. Using the UV–Vis spectra, optical properties including Fermi energy level, direct and indirect optical band gaps and Urbach energy were determined. The mentioned optical properties were calculated using Fermi–Dirac distribution function, Tauc's plots and absorption coefficient diagrams, respectively. Increment of optical properties with the increase of Y_2O_3 content (i.e. decrease of Fermi energy level from 3.59 to 3.17 eV, indirect band gap from 3 to 2.79 eV and direct band gap from 3.57 to 3.42 eV) confirms the better semiconducting behavior. Moreover, Urbach energy decreased from 0.27 to 0.23 eV arising from the network forming role of Y^{3+} ions in the glass network.

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1. Introduction

Among the variable glassy systems, an increasing interest has been devoted to the fluoride glasses because of their suitable optical properties, i.e., low phonon energy ($\sim 500\text{ cm}^{-1}$) and high transparency [1,2]. In contrary to their undeniable optical advantages, fluoride glasses have less favorable chemical, thermal and mechanical properties in parallel with fabrication problems [3]. On the other hand, oxide glasses e.g. silicate glasses are more mechanically, thermally and chemically stable [4]. In spite of the mentioned advantages, oxide glasses are inefficient in many optical applications due to their high phonon energies [4,5]. Since high optical properties are not sufficient for optical applications and stability is also demanded, there were attempts to combine advantages of both fluoride and oxide glasses. In that sense, glasses containing both fluorine and oxygen, called oxyfluoride glasses, have solved the problem to some extent [6]. In other words, introduction of oxygen to fluoride glass increases the stability and affects the glass formation [7]. Because of the appropriate electrical and optical properties, rare earth doped glasses have been paid much attention [8,9]. A considerable number of studies have been carried out on the effects of rare earth ions on the optical properties of oxyfluoride glass and glass ceramics like luminescence and upconversion behavior. To our knowledge, among the rare earth ions, although Y^{3+} ions have affected optical semiconducting behavior of other glass systems due to the surrounding ligand field of 4f-electron

of these ions [10,11], there was not any investigation of potential effects of them on oxyfluoride glasses.

Therefore, in the present study, oxyfluoride glasses doped with different amounts of Y_2O_3 were fabricated successfully. Corresponding optical values, i.e. Fermi energy level, direct and indirect optical band gaps and Urbach energy have been calculated using the UV–Vis spectra of them.

2. Experimental procedure

The designated compositions of glasses are presented in Table 1. Refining agents (As_2O_3 and Sb_2O_3) are also added to produce bubble-free samples. In order to prepare glass specimens, reagent grade materials were used. 50 g of the batches was melted in alumina crucibles at $1450\text{ }^\circ\text{C}$ for 1 h. The melts were poured on stainless steel molds preheated at $500\text{ }^\circ\text{C}$. With the aim of releasing the internal stress, glasses with the approximate thickness of 0.5 cm were annealed at $500\text{ }^\circ\text{C}$ for 30 min.

Standard principle of Archimedes was used to measure the density of specimens. Differential thermal analysis (DTA) curves of samples were plotted with the rate of $10\text{ }^\circ\text{C/min}$, using a DTG-60AH Shimadzu instrument, to measure crystallization peak temperatures (T_p). The Fourier transform infrared (FT-IR) transmittance spectra of the glass samples were measured by a Tensor 27 Bruker Company FT-IR spectrophotometer to study structural changes. Optical transmittance measurements were carried out in the UV–Vis spectrum range by means of a UV–Vis Shimadzu 1700 spectrophotometer instrument at room temperature. According to the results obtained from transmittance spectra, absorption and extinction coefficient, Fermi energy level, direct

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Table 1
Compositions of glasses with different amounts of Y_2O_3 .

Sample code	Composition (wt.%)					
	SiO ₂	Al ₂ O ₃	CaO	CaF ₂	Y ₂ O ₃	Sb ₂ O ₃
GY0.5	37.26	28.11	7.73	26.89	0.5	0.2
GY1	37.26	28.11	7.73	26.89	1	0.2
GY1.5	37.26	28.11	7.73	26.89	1.5	0.2

and indirect optical band gaps and Urbach energy of the samples were determined.

3. Results and discussion

3.1. Density and molar volume of samples

Density of the glassy samples was calculated using Eq. (1), which is derived from the standard principle of Archimedes.

$$D = \frac{W_1}{(W_1 - W_2)} \quad (1)$$

where W_1 and W_2 are the weights of samples in air and water, respectively.

According to the definition of density, molar volumes (V_m) of samples were obtained by applying Eq. (2).

$$V_m = \sum \left(\frac{M_i}{D} \right) \quad (2)$$

M_i , the molar mass of glass, is calculable from Eq. (3).

$$M_i = C_i A_i \quad (3)$$

where C_i and A_i are the molar concentration and the molecular weight of the i th component respectively [10,12].

Table 2 presents the results of density measurements for glasses doped with different amounts of Y_2O_3 . With the increase of Y_2O_3 content, an increasing trend of density is observed, followed by a decreasing V_m values. This outcome is justifiable by considering the network former role of Y_2O_3 . That is to say, Y^{3+} ions increase the bridging oxygens (BOs) by forming Y_2SiO_5 structure [10] and consequently with the emergence of more BOs, glassy network gets more compact (lower V_m). Therefore, due to the invert relation of density and molar volume, more Y_2O_3 percent lead to higher densities. These results can affect the optical properties.

3.2. Differential thermal analysis (DTA)

DTA curves with the heating rate of 10 ($^{\circ}C/min$) are presented in Fig. 1 for glasses with different amounts of Y_2O_3 . Appearance of two exothermic peaks in DTA results was expected on the basis of other previous reports [13–15]. The first peak is attributed to the crystallization of CaF_2 crystals. In contrast, there is no convincing interpretation for the second peak at about 900 $^{\circ}C$. As it is obvious in the DTA plots and listed in Table 2, the entrance of Y^{3+} ions into the glass network increases the crystallization peak temperature of CaF_2 from 690 to 719 $^{\circ}C$. Since crystallization mechanism of CaF_2 is a three-dimensional crystal growth

Table 2
Physical properties of oxyfluoride glasses containing different amounts of Y_2O_3 dopant.

Sample code	D (g/cm ³)	V_m (cm ³ /mol)	T_{P1} ($^{\circ}C$)	T_{P2} ($^{\circ}C$)
GY0.5	2.75	26.45	690	919
GY1	2.79	26.40	700	916
GY1.5	2.81	26.22	719	930

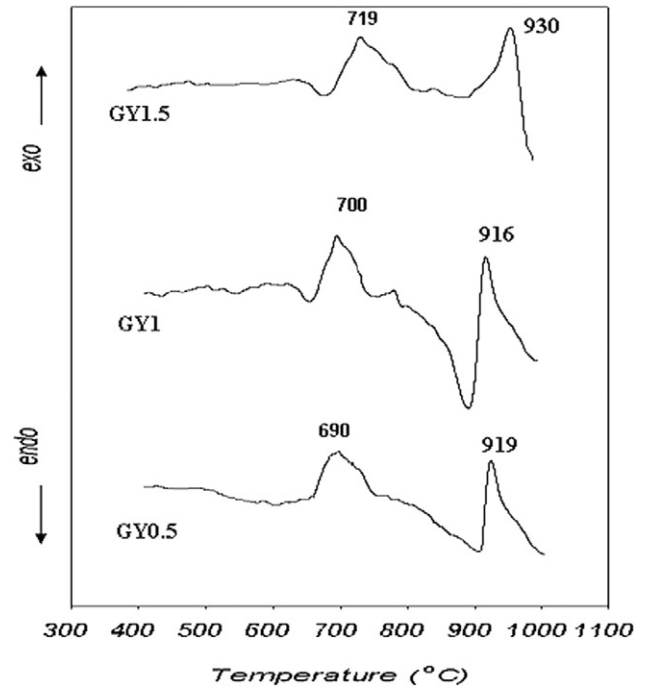


Fig. 1. DTA curves of the glass samples containing different amounts of Y_2O_3 .

process controlled by the diffusion [16], more BOs created by Y_2O_3 dopant decrease the mobility of ions. Thus, crystallization temperature shifts to higher temperature.

3.3. Fourier transform infrared (FT-IR) spectroscopy

For obtaining structural information of oxyfluoride glasses doped with various amount of Y_2O_3 , FT-IR spectroscopy has been carried out. Fig. 2 illustrates the FT-IR spectra of glasses extended from 4000 to 400 cm^{-1} . Using a significant amount of SiO₂ in the composition of

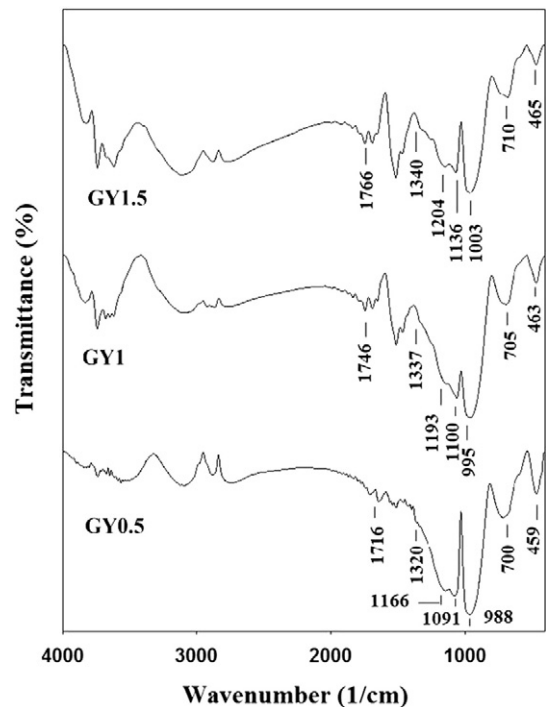


Fig. 2. FT-IR spectra of the oxyfluoride glasses containing different amounts of Y_2O_3 .

samples has resulted in bands related to the presence of SiO_4 tetrahedrons in the glass network. Bands placed in the ~ 460 , ~ 700 and $\sim 1170 \text{ cm}^{-1}$ are attributed to the rocking vibrations, symmetric stretching vibrations and asymmetric stretching vibrations of Si–O–Si bonds [17,18]. According to the network former role of Al^{3+} ions in the glass network, replacement of some Si^{4+} ions by Al^{3+} ions results in Al–O–Si bonds. Therefore, the absorption at $\sim 1090 \text{ cm}^{-1}$ is related to the asymmetric stretching vibration mode of Si–O–Al [19]. Although Ca–F bonds should have resulted in a peak at 443.30 cm^{-1} [20], overlapping with the rocking vibration mode of Si–O–Si has made this peak somehow difficult to distinguish. As Hill et al. [21] have reported, F^- ions associate with aluminum atoms in AlO_4 tetrahedra in oxyfluoride glasses. Hence, Al–F bonds create a peak at about 1350 and 1750 cm^{-1} in the FT-IR spectra [22]. On the other hand, due to the replacing of oxygens by F^- , Ca^{2+} ions attempt to compensate the negative charge induced in both $[\text{AlO}_4]^-$ and $[\text{AlO}_3\text{F}]^-$ tetrahedrons and create a band at $\sim 990 \text{ cm}^{-1}$ related to the Si–O–Ca bonds [17]. The peak observed at 3500 cm^{-1} may be emerged due to the O–H stretching of the surface water. The small peaks at about 1550 and 1610 cm^{-1} are related to the CO group and molecular water, respectively. The CO peak might be due to the residual carbonates introduced by the raw materials. As it is detectable from Fig. 2, more Y_2O_3 percent has shifted vibrations on the spectra to higher wavenumbers and decreased their peak intensity. These events are ascribed to the network forming role of Y^{3+} ions in the glass network and increase of BOs [10,23].

3.4. UV–Vis absorption spectra and evaluation of optical constants

3.4.1. Absorption and extinction coefficients

The Beer–Lambert law explains the relationship between the absorption of light and the properties of the material through which the light is traveling. The general Beer–Lambert law is written as:

$$A = \alpha bc \quad (4)$$

where A is the measured absorbance, α is the wavelength-dependence absorption coefficient, b is the path length and c is the concentration.

On the other hand, transmittance (T) which is usually measured during experiments is defined as:

$$T = I/I_0 \quad (5)$$

where I is the intensity of light after passing through the sample and I_0 is the initial light intensity. The relation between absorbance and transmittance is:

$$A = -\ln T = -\ln(I/I_0). \quad (6)$$

Therefore, one can rewrite the Beer–Lambert law as:

$$T = \exp(-\alpha bc). \quad (7)$$

For transparent glasses, the Beer–Lambert law is given by:

$$I = I_0 \exp(-\alpha b). \quad (8)$$

Consequently, the absorption coefficient of such glasses is computable, provided that the transmittance and the thickness of specimens are given [24,25].

In addition, the extinction coefficient (K) of a material, i.e., the imaginary part of the complex index of refraction, is determinable by using Eq. (9):

$$K = \frac{\alpha \lambda}{4\pi} \quad (9)$$

where α is the absorption coefficient and λ is the wavelength of incident photon to sample in the vacuum [26].

In order to calculate the optical constants, the UV–Vis spectra of samples were recorded and presented in Fig. 3. The UV–Vis absorption spectra depict that the absorption has increased along with the increase of Y_2O_3 content. One can conclude that the creation of more BOs in the glass structure somewhat prohibits the optical transmission.

3.4.2. Determination of Fermi energy level

According to the strong absorption of UV absorption bonds, extinction coefficient (K) obeys from the Fermi–Dirac distribution function, thus, the Fermi energy level can be calculated applying Eq. (10):

$$K(\lambda) = \frac{1}{1 + \exp\left(\frac{E_f - E}{K_B T}\right)} \quad (10)$$

where E_f is the Fermi energy, E is the variable photon energy probing the sample k_B is the Boltzmann constant and T is the ambient temperature (297 K) [27]. Fig. 4 depicts K vs. $h\nu$ plots for different glass samples. The calculated values for Fermi energy of samples are presented in Table 3, which are achieved from least square fittings of Eq. (10). Reduction of Fermi energy level in the presence of more Y_2O_3 indicates a change to better semiconducting properties.

3.4.3. Direct and indirect optical band gaps

According to Mott and Davis [28], the absorption of light by amorphous solid depends on the energy ($h\nu$) of the incident photon and E_g^{opt} is the optical band gap of the sample. This behavior may be represented by:

$$\alpha = \beta^2 \frac{(h\nu - E_g^{opt})^n}{h\nu} \quad (11)$$

where β is a constant related to the extent of the band tailing, n is the index which can have different values 2, 3, 1/2 and 1/3 corresponding to indirect allowed, indirect forbidden, direct allowed and direct forbidden transitions, respectively [10]. Therefore, by plot $(\alpha h\nu)^{1/n}$ vs. photon energy (Tauc's plots), the intercept of the obtained line divided by slope is equal to the energy band gap of optical transitions [29]. Fig. 5 illustrates the Tauc's plots of the glasses and Table 3 shows the direct and indirect allowed optical band gaps of the glasses calculated from plots of Fig. 5. As discussed in previous sections, introduction of Y_2O_3 has decreased the number of non-bridging oxygens (NBOs), which causes the reduction of average band gap energy and results in the fall of conduction band's level. Therefore, a systematic decrease of the optical band gap has been established in more Y_2O_3 containing glasses [10, 27]. On the other hand, it is reported that Y^{3+} ions may cause the creation of new energy levels in the forbidden band gap which reduce the band gap energies [30,31].

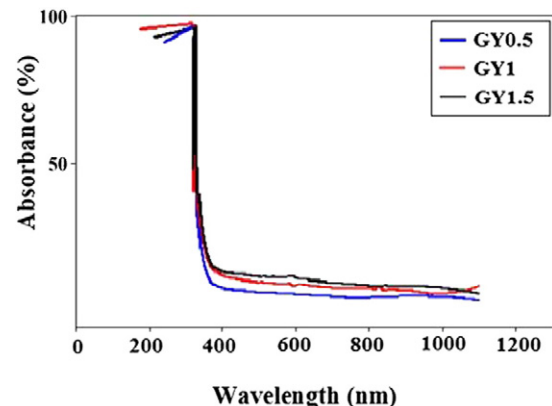


Fig. 3. UV–Vis spectra of glass specimens in the presence of various amounts of Y_2O_3 .

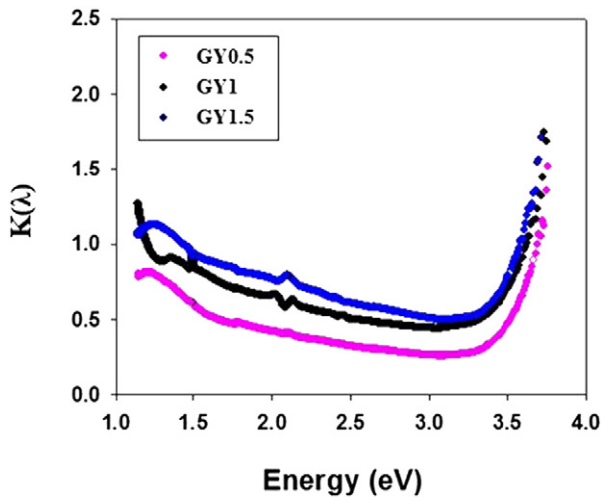


Fig. 4. Extinction coefficient vs. energy plots of glasses containing different amounts of Y_2O_3 .

3.4.4. Urbach energy

The optical absorption in amorphous semiconductors near the absorption edge is usually characterized by three types of optical transitions corresponding to transitions between tail and tail states, tail and extended states, and extended and extended states. The first two types correspond to $h\nu \leq E_g^{opt}$ and the third one corresponds to $h\nu \geq E_g^{opt}$. Thus, the plot of absorption coefficient versus photon energy (α vs. $h\nu$) has three different regions. In the second region, the absorptions is related to transitions from the localized tail states above the valence band edge to extended states in the conduction band and/or from extended states in the valence band to localized tail states below the conduction band. The spectral dependence of absorption coefficient usually follows the so-called Urbach rule (Eq. (12)):

$$\alpha = \beta \exp\left(\frac{h\nu}{E_U}\right). \quad (12)$$

For many amorphous semiconductors, E_U has been related to the width of the valence (or conduction) band tail states and degree of disorder [32,33]. Table 3 represents the Urbach energy values of the glasses calculated from the least square fitting of $\ln \alpha$ against photon energy plots in the tailing part of localized states and listed in Table 3. With increasing the Y_2O_3 content, the Urbach energy of glasses has been decreased. This outcome can be related to a decrease in the broadening due to static disorder-related part [27]. In addition as it is mentioned in previous sections, more Y_2O_3 additive result in more BOs, which is reported as a reason of lower Urbach energy [10,11].

4. Conclusion

In summary, Y^{3+} ions doped oxyfluoride glasses were prepared by means of the common melting process. Decreasing the crystallization temperature of CaF_2 and enhancement of density in glasses with higher content of Y_2O_3 implicitly proved the network modifier role of Y^{3+} ions. FT-IR spectroscopy also confirmed the creation of BOs, i.e., characteristic

Table 3
Optical properties of oxyfluoride glasses with different amounts of Y_2O_3 .

Sample	Energy (ev)			
	E_f	$E_{indirect}$	E_{direct}	E_U
GY0.5	3.59	3.00	3.57	0.27
GY1	3.43	2.82	3.52	0.24
GY1.5	3.17	2.79	3.42	0.23

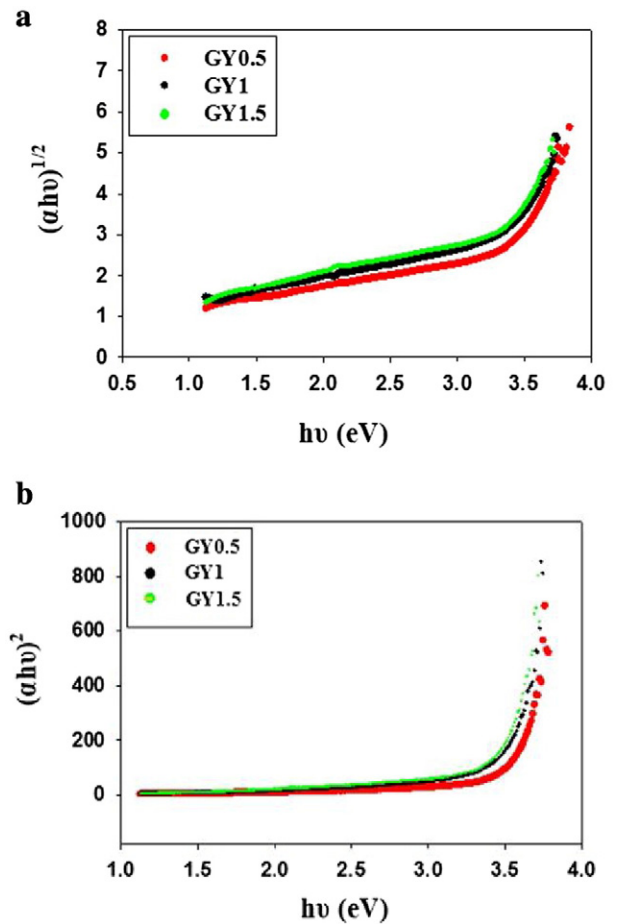


Fig. 5. Tauc's plots for (a) indirect and (b) direct band gaps.

features were shifted to lower wavenumbers and their intensities were decreased. The optical band gap and Fermi energy level of the glasses were decreased, that is to say, their semiconductor characteristics were increased. Urbach energy was also decreased for the glasses with the higher Y_2O_3 contents. In addition, the degree of crystallinity and order was increased as an outcome of Urbach energy reduction.

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